

# Falcon Service Manual (3rd Edition)

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Area: manuals

## 1 Engine

Service coverage for the Cascadia Falcon 2.0T and 1.6 engines: driveability diagnosis, ignition service, and related emissions interactions.

### 1.1 Misfire Diagnosis

Misfire codes P0300 (random/multiple) and P0301 (cylinder 1) on the 2.0T are most often ignition-related. Confirm fuel trims and compression are within range before condemning a coil. Begin with the Ignition System checks. If a coil has already been replaced and the misfire returns, the original part may be at fault — see Revised Ignition Coil for Repeated Misfire . A persistent misfire can also set catalyst code P0420 downstream; see Catalyst Efficiency .

### 1.2 Ignition System

The 2.0T uses a coil-on-plug ignition system. Inspect coil boots and connectors for heat damage before parts replacement.

#### 1.2.1 Ignition Coil Replacement

Replace the ignition coil IC-2042-A as a complete set; mixing old and new coils masks a weak cylinder. Torque the hold-down bolts to 9 Nm. On early-production vehicles the connector can overheat — refer to Safety Recall RC-2021-04 and install the revised coil IC-2042-B. Always replace the spark plug set when servicing coils (see Spark Plug Service ).

#### 1.2.2 Spark Plug Service

Replace spark plug set SP-1108 at the specified interval or whenever the coils are serviced. Gap to 0.7 mm. Worn plugs are a common contributor to P0301.

## 2 Emissions

Catalyst monitoring and engine-control software for the Falcon platform.

### 2.1 Catalyst Efficiency

Code P0420 indicates catalyst efficiency below threshold. Rule out upstream misfire (see Misfire Diagnosis ) before replacing the catalytic converter CAT-5500, as raw fuel from a misfire damages the catalyst.

### 2.2 ECM Software Updates

Some driveability and emissions complaints are resolved by an ECM-9901 software update rather than hardware. See Updated Catalyst Monitor Calibration .

## 3 Brakes

Front brake service and hydraulic system for the Falcon platform.

### 3.1 Front Brake Service

Service the front brake pad kit BP-7720 and rotor pair BR-7731 as a set. Machining is not approved on this rotor. For brake judder after a pad change, see Brake Judder After Front Pad Replacement .

### 3.2 Brake Hose Replacement

Replace the front brake hose BH-7745 if cracked or swollen. Bleed the system in the specified sequence after service.

## **4 ABS**

Wheel-speed and anti-lock control diagnosis.

### **4.1 Wheel Speed Sensors**

Code C0035 indicates a left front wheel speed sensor circuit fault. Inspect the sensor WSS-3300 and its harness routing near the brake hose. Sensor faults can cascade to ABS communication loss; see ABS Module Diagnosis .

### **4.2 ABS Module Diagnosis**

Code U0121 indicates lost communication with the ABS control module. Check power, ground, and CAN continuity before module replacement. A recall applies to certain harness routings — see Wheel Speed Sensor Harness Chafing .

## **5 Electrical**

Charging and starting system service.

### **5.1 Charging System**

A no-charge condition sets P0562 (system voltage low). Test the alternator ALT-8810 output before replacement and confirm the battery is healthy (see Battery Service ).

### **5.2 Battery Service**

Replace the AGM battery BAT-1200 with the same chemistry; register the new battery to the charging system where required.